

PROG3240-26W Business Intelligence: Final Project Assignment 3

Assignment Info: (30% of total grade)

Due Date: Sunday, Apr 5th @11:59 PM

There will be strict deadlines for the assignment. 15% per day will be subtracted for a late submission, to a maximum of 2 days late.

This is a group assignment (Min 2 and Max 3 students). Significant overlap between submissions from two or more students may be flagged for plagiarism.

Grading:

Item	Description	Weight
Part 1:		10
Requirement 1	- Dimensional Model tables (including PKs, FKs, and Indexes).	5
Requirement 2	- Date dimension & Stored Procedure to load it	3
Requirement 3	- Create Compelling Warehouse Query.	2
Part 2:		20
Requirement 4	- Extracts T-SQL, Python, SSIS.	6
Requirement 5	- Transforms T-SQL, Python, SSIS.	8
Requirement 6	- Create T-SQL ETL Loads	4
Requirement 7	- Load rest of data to DWH and Query	2
Submission Requirements	Penalty for failure to adhere to the submission requirements and errors during the execution	Up to -10%
Total		30/30

Initialization:

This assignment must run against the database WideWorldImporters used throughout this course.

Business Requirements:

Part 1: You have been asked to create a star schema dimensional model Data Warehouse/Data Mart with these business requirements (updated from our Class notes as discussed in class):

Requirement 1 – Dimensional Model tables (5 Marks) (Add Supplier Dimension to the Star Schema)

You are to continue working for a company **updating** the **current WWI DataMart SQL (class notes)**. You already have this Dimensional Star Schema database warehouse build, which you will be **ADDING/Updating** to add information about Suppliers. The business has been ordering products from different suppliers

So, In addition to the current solution that includes:

*We would like to track **sales** information for WideWorldImporters. We would like to analyze which **products** are being ordered to determine how things like **brand, colour, and price** may impact gross **sales**. We would like to identify preferences for specific **customers** so that we may more effectively predict which **products** they may be interested in. Similarly, we would like to be able to determine if certain **cities** have **brand, colour, and price***

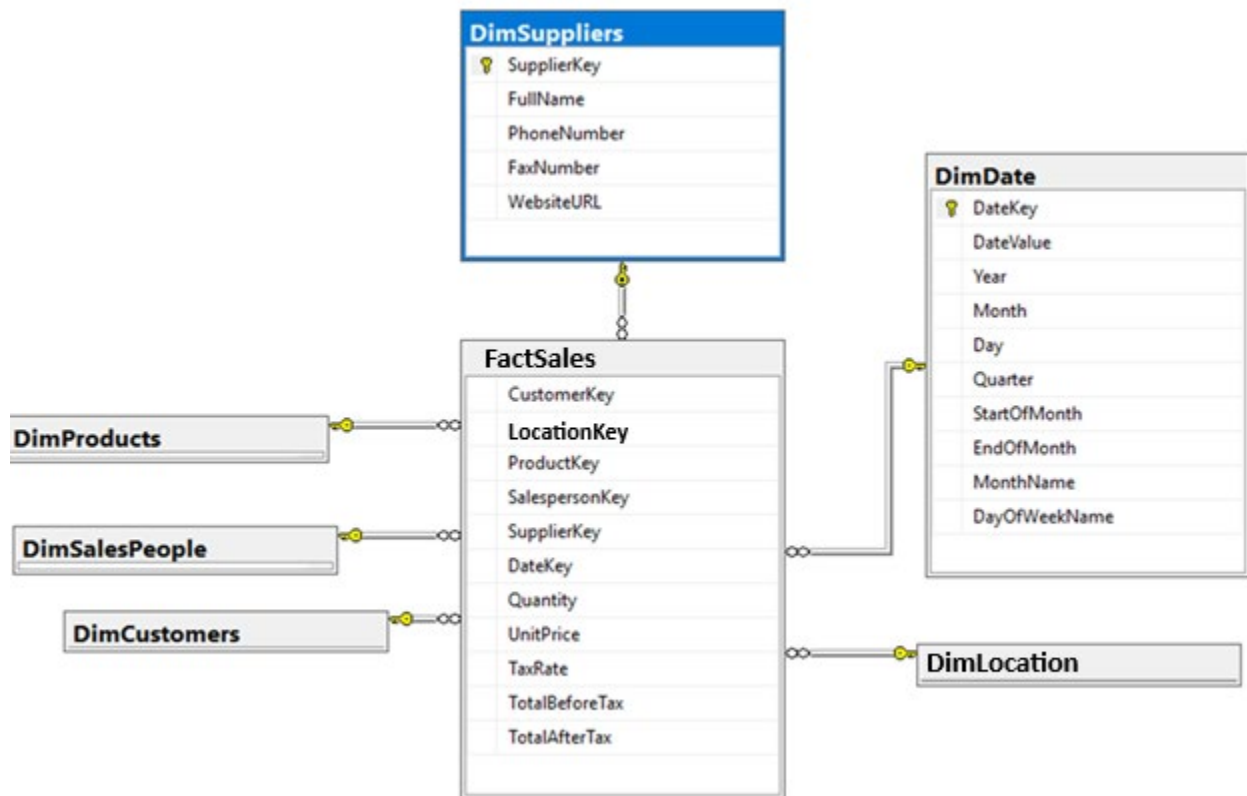
preferences. We would also like to determine whether certain **salespeople** have more success selling certain **products**, selling to certain **customers**, or selling in certain **cities**. We would also like to determine whether certain **suppliers** **products** show more successful sales, and which **customers** and **cities** they are most successful in. We need to track the **date** of each **order** to be able to track the impact of our change over **time**.

... you also are required to track / be able to analyse historical data to :

This leaves us with the following distinct concepts: **sale; product; brand; colour; price; customer; city; salesperson; supplier; date; order; time.**

... As a result of those additional requirements & thru discussion with the business experts, you have determined that this is the expected NEW structure of the Updated/Complete Solution:

This leads us to develop/build this structure:



Using the SQL already provided in the class notes, and adding on the additional dimTable as per the model diagram above, build & execute the SQL to build the tables required for this data warehouse star schema model. Review the source of the Supplier table to determine the unique/business key similar to how the customer dimension key is determined. Also, for this Supplier dimension, assume that for the remaining non-key attributes, if they change, they would impact associated facts, so set the SCD level appropriately. Since data warehouse users will usually analyse their business by seeking records in dimension table(s), and then seek their corresponding/matching records in the related fact table, build the appropriate indexes to ensure this work is optimized.

Update your SQL solution (based on starting with the Initial database that already has the other Dim Tables & the FactSales tables & relationships created) to include the new DimSupplier dimensional table, and update any related fields/constraints/Indexes in any related tables.

Requirement 2 – Date dimension & Stored Procedure to load it (3 Marks)

Create a stored procedure to insert into the date dimension table, using a single DateValue parameter as an input DATE field. Use While Loop to add 5 years date values (starts from CY2012)

Requirement 3 – Create Compelling Warehouse Query (2 Marks)

Write a query that will return Customer, City, Salespeople, Products, suppliers, and dates for Order facts using this dimensional table... (You decide on the business scenario & fields to provide the solution for your query... marks will be 3 for a working query, and 3 more for particularly compelling queries... Think “Predict the **Future**”)

Part 2: This part will require creating a significant amount of stored procedure SQL. (Exclude DimPickingStaff)

Requirement 4 –Extracts (6 Marks) (at least 1 python and 1 SSIS package)

Create stage tables to insert the extracted data into, and Write stored procedures that will obtain all the required data from the following source data sets from WideWorldImporters:

- Customers – Query that joins Customers, CustomerCategories, Cities, StateProvinces, and Countries.
- Products – Query that joins StockItems and Colours
- Salespeople – Query of People where IsSalesperson is 1
- Orders – Query that joins Orders, OrderLines, Customers, and People, and accepts an @OrderDate as a parameter, and only selects records that match that date.
- Suppliers – Query that joins Suppliers and SupplierCategories (Business Analyst and SME review of the Supplier source tables suggests that SupplierCategory might also influence sales orders, so please add the SupplierCategoryName field to the appropriate table in the dimensional model).

Test each of your Extract stored procedures by executing each one of them. When testing the orders extract, use ‘2013-01-01’ as the date.

Requirement 5 – Transforms (8 Marks) (at least 1 python and 1 SSIS package)

Create the PreLoad staging tables that match the structure of the destination (dim) tables, and Stored Procedures to perform the transformations of the source to destination data. Include in the transformations:

- 1) Updating any SCD’s that require it.
- 2) For SCD 1 dimensions, create and use a Sequence to set the dimension business key to the NEXT VALUE for any newly added records (ie. did NOT EXIST before), and use the existing surrogate key if a record already exists when updating it from the stage table.
- 3) For SCD 2 dimensions, add updated records, add existing records and expire as necessary, create newly added records (ie. Did NOT EXIST before), and expire missing records. Reference this text from the class notes : We are comparing our extracted records against the current records in our data mart table:
 - a. When there is a match, but **no non-key attributes have changed**, we **add the record as-is** so it is available for our fact transform.
 - b. When there is a match and a **non-key attribute has changed**, we **create a new record** and **expire the previous record**.
 - c. When there is **no match in the data mart**, we **create a new record**.
 - d. When there is **no match in our extract** for a record in the data mart, we **expire the existing record**.

- e. make the **new DimSupplier tables** based on this dimension **a SCD type 2**, including updating the associated Stored Procs. Note you should be able to use some of the existing SQL code that is already created/functional from the supplied SQL.
- 4) A simple validation error if there are no records found, where appropriate.
- For our orders fact transform, use the staged data to look up our surrogate keys by business key, and follow the same aggregation of the measure data as we did in our class notes (realizing that our BA and/or SME might add/change these requirements depending on what business conditions they are trying to understand/predict.)

Requirement 6 – Create ETL Loads (4 Marks)

Create stored procedures that will load the dimension tables with any changed records and load the fact table in the WideWorldImporters Data mart/Data warehouse. Remember to follow the best practices for creating stored procedures that ensures that if part of the updates for a table fail, the rest of the updates to the table will be rolled back.

Requirement 7 – Load rest of data to DWH and Query (2 Marks)

Execute the above procedures for 4 days worth of Orders to load the Data Mart (2013-01-01 to 2013-01-04).

Run the query that you created in Requirement 3 to return Customer, City, Salespeople, Products, suppliers, and dates for Order facts using this dimensional table... (You decide on the business scenario & fields to provide the solution for your query... Remember you are expecting to create compelling queries... ie. What would an expert business person like to learn that would help them to “Predict the **Future** “ & *then USE that information to make their business better!*)

Remember to test out your SQL script & note that your professor will only be running it all in 1 Script / from the TOP & will not be able to analyse/diagnose/fix any of your SQL errors (so please comment out any parts of your script that are generating errors – I can give partial marks ...).

Submission Requirements:

Make sure you are satisfied with the database you have created before scripting it out.

- 1) Each requirement will be graded independently, though most requirements are at least partially dependent on the successful completion of previous steps.
- 2) Submit a single “Part1_YourGroup#.sql” file for Part 1.
- 3) Submit “Part2_YourGroup#.sql”, “Part2_YourGroup#.py”, “Part2_YourGroup#.dtsx” files for Part 2
- 4) Use comments to clarify where your work for a requirement begins and ends. For Example:

```
/* REQUIREMENT 1 */
SELECT *
FROM JustAnExample;

/* REQUIREMENT 2 */
SELECT *
FROM AlsoAnExample;
```

- 5) You are encouraged to test your code before submitting. Your script must execute without error against WideWorldImporter and WWI_DM databases. **DO NOT ZIP/COMPRESSED YOUR FILES**